Teacher answer key: Primary 5, Semester 1 (DRAFT)

DRAFT: this answer key is for owner review. Do not give it to pupils. Contact: pypteltd@gmail.com (business email for now, to be changed later)

This key lists the answers and the result of the check on every page. You do not need it for routine marking: a pupil who has a working check has the right answers. To spot-check, compare the check strip with the line "Check result" below.

How the topic tags work

Each entry names the syllabus outcomes the puzzle practises, using the outcome numbers of the Primary 5 (Standard) content in the Mathematics Syllabus, Primary (2021 edition, pages 41 and 42) where our summary prints them. The wording of each outcome line is our own summary, not the syllabus text. The references are for your planning only and are not printed on the pupil pages. "Problem solving" marks a page whose questions come from the kinds found in school tests rather than from one numbered outcome.

Pages in this booklet (Terms 1 and 2)

Strand	Pages
Number and Algebra - Whole Numbers	S1-01, S1-02, S1-03, S1-04, S1-05, S1-06, S1-07
Number and Algebra - Fractions	S1-08, S1-09, S1-10, S1-11, S1-12, S1-13, S1-14, S1-15, S1-16
Number and Algebra - Decimals	S1-17, S1-18, S1-19, S1-20

The page code in the footer, for example S1-07 (P5-07), gives the page in this booklet and, in brackets, its number in the full set of 40.

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S1-01 (P5-01) Millions 1 star

Topic: Whole numbers: numbers to 10 million

Syllabus outcomes: Whole Numbers: numbers to 10 million. Notes on the tag: numbers to 10 million: reading and writing numbers in numerals and words, place value
Puzzle type: Lettered choices (place value). How the pupil checks: Word from the letters of the right choices (6 letters).
Answers: 1) 5000; 2) four million, five thousand and eleven; 3) 9 060 005; 4) 4 020 401; 5) 1 552 683; 6) 6 187 282
Check result: Word: FLIGHT
Teacher note: Six place-value questions with numbers to 10 million: the value of a digit, a numeral in words, words as a numeral, an expanded number, the greatest of three and the number with a digit in a named place. The wrong choices are the usual slips (a place value one place too high or too low, two neighbouring digits swapped, a zero in the wrong place). Each choice has a letter; the right choices spell the word. Every other way of choosing was tested against a very large word list. A box left blank gives no letter and the check cannot see it.

S1-02 (P5-02) Seven Digits 2 stars

Topic: Whole numbers: making numbers to 10 million from digit cards

Syllabus outcomes: Whole Numbers: numbers to 10 million. Notes on the tag: numbers to 10 million: place value, reading and writing numbers
Puzzle type: Code-breaker (digit cards). How the pupil checks: Code word (6 letters).
Answers: 1) 2 056 789; 2) 3 045 789; 3) 9 867 410; 4) 9 865 310; 5) 1 023 457; 6) 9 765 430
Check result: Code word: FROZEN
Teacher note: Six digit-card questions, each with seven different digit cards including a zero. The pupil makes the smallest or greatest number, the greatest even or smallest odd number, or the smallest or greatest number with a stated digit in the thousands or ten thousands place. A number cannot start with 0. The decoy letters stand for the answers a pupil gets by putting the cards in the wrong order, by forgetting the zero rule, or by answering a different question on the same cards. A box left blank gives no letter and the check cannot see it.

S1-03 (P5-03) Zero Hunt 1 star

Topic: Whole numbers: multiplying and dividing by 10, 100 and 1000

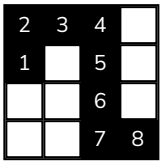
Syllabus outcomes: Whole Numbers: multiply and divide by 10, 100, 1000 and multiples. Notes on the tag: multiplying and dividing whole numbers by 10, 100 and 1000
Puzzle type: Ordering (lettered cards). How the pupil checks: Letters of the cards in the order asked spell a word (6 letters).
Answers: Order (smallest first): 9000 ÷ 1000 = 9, 2500 ÷ 100 = 25, 380 ÷ 10 = 38, 26 000 ÷ 100 = 260, 9986 × 10 = 99 860, 6772 × 100 = 677 200
Check result: Word: BACKUP
Teacher note: Six cards, each a whole number multiplied or divided by 10, 100 or 1000. The cards are not in the order of the numbers printed on them: the pupil has to work out each answer and then order the six answers from smallest to greatest. The letters, read in that order, spell a word with exactly one arrangement that is a word, so no wrong order of the six cards spells a word (every wrong order was tested against a very large word list). A box left blank gives no letter and the check cannot see it.

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S1-04 (P5-04) Multiples Route 2 stars

Topic: Whole numbers: multiplying and dividing by multiples of 10, 100 and 1000

Syllabus outcomes: Whole Numbers: multiply and divide by 10, 100, 1000 and multiples. Notes on the tag: multiplying and dividing whole numbers by 10, 100, 1000 and their multiples
Puzzle type: Maze / path. How the pupil checks: Route: the line must reach FINISH through the right number of boxes, every box following the rule.
Answers: Route (row, column from top-left): (2,1) 2 880 000 ÷ 300 -> (1,1) 576 000 ÷ 60 -> (1,2) 8 640 000 ÷ 900 -> (1,3) 864 000 ÷ 90 -> (2,3) 4 800 000 ÷ 500 -> (3,3) 192 × 50 -> (4,3) 384 000 ÷ 40 -> (4,4) 768 000 ÷ 80
Check result: Route reaches FINISH through 8 boxes; every box on it follows the rule.

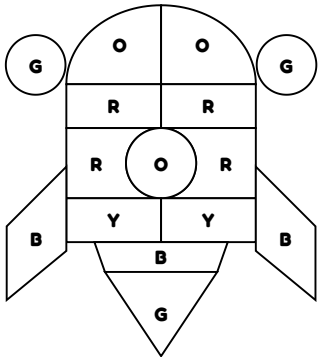


Teacher note: A 4 by 4 grid of sixteen calculations, each a whole number multiplied or divided by a multiple of 10, 100 or 1000 . The boxes off the route make answers that come from a slip in the zeros (ten or a hundred times too big or too small) and some other near misses; some extra boxes that do make 9600 are dead ends. Counting the boxes shows a line that takes a shortcut.

S1-05 (P5-05) Colour the Rocket 1 star

Topic: Whole numbers: order of operations and brackets

Syllabus outcomes: Whole Numbers 2.2, Whole Numbers 2.3. Notes on the tag: 2.2, 2.3 (order of operations and use of brackets, without a calculator)
Puzzle type: Colour by code (pupils colour with their own crayons). How the pupil checks: Colour key and tally: every answer must be on the key, the number of parts of each colour must match the box, and the parts in the same place on the left and on the right match.
Answers: Colour key: 12 = yellow, 18 = red, 24 = blue, 30 = green, 36 = orange
Check result: Tally: yellow 2, red 4, blue 3, green 3, orange 3; the parts in the same place on the left and on the right match.



Teacher note: Every part is a calculation with the four operations, with or without brackets: multiplication and division come before addition and subtraction, and brackets come first. Pupils colour the black-and-white outline with their own crayons (or, for a pencil-only copy, write Y, R, B, G or O in each part: the tally and the matching parts can be checked the same way). Check strip: count the parts of each colour (the tally is printed); the parts in the same place on the left and on the right match.

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S1-06 (P5-06) Brackets Match 2 stars

Topic: Whole numbers: order of operations and brackets

Syllabus outcomes: Whole Numbers 2.2, Whole Numbers 2.3. Notes on the tag: order of operations and use of brackets, without a calculator
Puzzle type: Matching (lettered cards). How the pupil checks: Letters of the matching cards, tile 1 to 6, spell a word.
Answers: C: (34 − 16) × 4 ÷ 3 = 24; R: 42 ÷ (8 + 6) × 9 = 27; A: (56 − 6) ÷ 2 + 4 = 29; F: 28 + 4 × (9 − 2) = 56; T: (39 + 11) ÷ 5 × 7 = 70; S: 49 × (3 + 3) − 4 = 290
Check result: Word: CRAFTS
Teacher note: Six cards with brackets and the four operations; the value of each is a whole number found in the usual order (brackets first, then multiplication and division from left to right, then addition and subtraction). Every card gives a different value when the brackets are ignored, so a pupil who ignores them finds no matching tile or the wrong one. The tiles are in order of their values, smallest first; the letters, read from tile 1 to tile 6, spell a word with exactly one arrangement that is a word, so no wrong matching spells a word. A box left blank gives no letter and the check cannot see it.

S1-07 (P5-07) Both Sides Equal 2 stars

Topic: Whole numbers: order of operations and brackets in number sentences

Syllabus outcomes: Whole Numbers 2.2, Whole Numbers 2.3 (problem solving). Notes on the tag: order of operations and use of brackets: finding the missing number
Puzzle type: Balance (put the answer back). How the pupil checks: Put every number back in its box: both sides of every line must be equal.
Answers: 1) 10; 2) 88; 3) 21; 4) 184; 5) 7; 6) 14
Check result: Every line is true when its number is put back.
Teacher note: Six number sentences, each with one empty box and brackets, multiplication, division, addition and subtraction. The pupil finds the number by working backwards or by trying numbers, then puts it back in the box to see whether both sides are equal. Each sentence has one answer. The page has no code word: a wrong number makes its line false, and a blank box is seen at once.

S1-08 (P5-08) Sharing Equally 1 star

Topic: Fractions: dividing a whole number by a whole number

Syllabus outcomes: Fractions 1.1. Notes on the tag: dividing a whole number by a whole number: the quotient as a fraction or a mixed number
Puzzle type: Code-breaker (division as a fraction). How the pupil checks: Code word (6 letters).
Answers: 1) 1/4; 2) 1/5; 3) 2 3/4; 4) 1 5/6; 5) 7/8; 6) 1 4/5
Check result: Code word: DEVOUR
Teacher note: Six divisions where the quotient is a fraction or a mixed number: four bare divisions (two that need simplifying) and two sharing stories. The decoy letters stand for the usual slips: the fraction turned upside down, the whole number part only, or the remainder only. A box left blank gives no letter and the check cannot see it.

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S1-09 (P5-09) Fractions as Decimals 1 star

Topic: Fractions: expressing fractions as decimals

Syllabus outcomes: Fractions 1.2. Notes on the tag: expressing fractions as decimals
Puzzle type: Matching (lettered cards). How the pupil checks: Letters of the matching cards, tile 1 to 6, spell a word.
Answers: B: 7/50 = 0.14; E: 3/20 = 0.15; H: 5/8 = 0.625; I: 7/8 = 0.875; N: 1 1/5 = 1.2; D: 2 7/20 = 2.35
Check result: Word: BEHIND
Teacher note: Six cards with fractions and mixed numbers (denominators from 2 to 50) and six tiles with decimals of one, two or three places, in order of size. A pupil finds each decimal by dividing the numerator by the denominator or by making an equivalent fraction with denominator 10, 100 or 1000. The letters, read from tile 1 to tile 6, spell a word with exactly one arrangement that is a word, so no wrong matching spells a word. A box left blank gives no letter and the check cannot see it.

S1-10 (P5-10) Mixed Number Magic 2 stars

Topic: Fractions: adding and subtracting mixed numbers

Syllabus outcomes: Fractions 2.1. Notes on the tag: adding and subtracting mixed numbers
Puzzle type: Magic square (mixed numbers). How the pupil checks: Total: all eight lines of the square must make the printed total.
Answers: Empty squares, row by row: 2 1/4, 9 3/4, 3 1/2, 12 1/4, 6. Full square: 8 1/2 2 1/4 11 / 9 3/4 7 1/4 4 3/4 / 3 1/2 12 1/4 6
Check result: Every line makes 21 3/4.

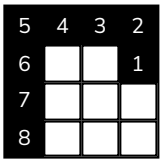
8 1/2	**2 1/4**	**11**
9 3/4	**7 1/4**	**4 3/4**
3 1/2	**12 1/4**	**6**

Teacher note: A 3 by 3 magic square with mixed numbers in quarters and halves (or in thirds). Four squares are given and the total of every line is printed. Each empty square is found by subtracting the numbers already in a line from the total, one line at a time; every empty square can be reached this way. The check is the total: a wrong number makes at least one of the eight lines wrong. All numbers in the square are different.

S1-11 (P5-11) Mixed Number Route 1 star

Topic: Fractions: adding and subtracting mixed numbers

Syllabus outcomes: Fractions 2.1. Notes on the tag: adding and subtracting mixed numbers
Puzzle type: Maze / path. How the pupil checks: Route: the line must reach FINISH through the right number of boxes, every box following the rule.
Answers: Route (row, column from top-left): (2,4) ⟨5 5|12⟩ – ⟨1 2|3⟩ -> (1,4) ⟨1|2⟩ + ⟨3 1|4⟩ -> (1,3) ⟨2 1|2⟩ + ⟨1 1|4⟩ -> (1,2) ⟨2 2|3⟩ + ⟨1 1|12⟩ -> (1,1) ⟨1 5|6⟩ + ⟨1 11|12⟩ -> (2,1) ⟨6 1|12⟩ – ⟨2 1|3⟩ -> (3,1) 5 – ⟨1 1|4⟩ -> (4,1) 7 – ⟨3 1|4⟩
Check result: Route reaches FINISH through 8 boxes; every box on it follows the rule.



Teacher note: A 4 by 4 grid of sixteen additions and subtractions of mixed numbers (denominators 2, 3, 4, 6 and 8, never more than two different ones in one box). The boxes off the route make answers a fraction or a whole away from 3 3/4; some extra boxes that do make 3 3/4 are dead ends. Counting the boxes shows a line that takes a shortcut.

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S1-12 (P5-12) Fraction Chain 1 star

Topic: Fractions: multiplying a fraction by a whole number

Syllabus outcomes: Fractions 2.2. Notes on the tag: multiplying a proper or improper fraction by a whole number
Puzzle type: Number chain (loop). How the pupil checks: Loop: following the arrows round, the last arrow must give the start number again.
Answers: Circles in order: 48, 40, 70, 77, 35, 56; last arrow x 6/7 returns to 48
Check result: The loop closes at the start number.
Teacher note: A ring of six circles joined by arrows. The start number is printed; each arrow multiplies the number in one circle by a fraction (some fractions are greater than 1) and gives a whole number in the next circle. The pupil fills five circles. The last arrow must bring the chain back to the start number: a wrong number in any circle carries on round the loop and the chain does not close. Two slips that cancel out are not detected.

S1-13 (P5-13) Fraction of a Quantity 1 star

Topic: Fractions: multiplying a fraction by a whole number

Syllabus outcomes: Fractions 2.2 (problem solving). Notes on the tag: multiplying a proper or improper fraction by a whole number (a fraction of a quantity)
Puzzle type: Code-breaker (fractions of quantities). How the pupil checks: Code word (6 letters).
Answers: 1) 28; 2) 30; 3) 14; 4) 63; 5) 25; 6) 12
Check result: Code word: PEANUT
Teacher note: Six fractions of a quantity, in a bare question and in short stories (length, money, a class, water, time). One fraction is greater than 1. The decoy letters stand for the usual slips: dividing by the denominator only, multiplying by the numerator only, or taking the answer away from the whole. Every quantity divides exactly by the denominator. A box left blank gives no letter and the check cannot see it.

S1-14 (P5-14) Fraction Products 2 stars

Topic: Fractions: multiplying a proper fraction by a proper or improper fraction

Syllabus outcomes: Fractions 2.3. Notes on the tag: multiplying a proper fraction by a proper or improper fraction
Puzzle type: Ordering (lettered cards). How the pupil checks: Letters of the cards in the order asked spell a word (6 letters).
Answers: Order (smallest first): ⟨3|10⟩ × ⟨1|3⟩ = 1/10, ⟨1|3⟩ × ⟨5|9⟩ = 5/27, ⟨4|9⟩ × ⟨6|7⟩ = 8/21, ⟨7|10⟩ × ⟨9|10⟩ = 63/100, ⟨5|9⟩ × ⟨9|7⟩ = 5/7, ⟨3|4⟩ × ⟨11|8⟩ = 1 1/32
Check result: Word: MELODY
Teacher note: Six cards, each the product of two fractions: a proper fraction times a proper fraction (four cards) or times an improper fraction (two cards). The cards are not in order: the pupil works out each product and puts the six products in order from smallest to greatest. The letters, read in that order, spell a word with exactly one arrangement that is a word, so no wrong order spells a word. A box left blank gives no letter and the check cannot see it.

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S1-15 (P5-15) Big Products 2 stars

Topic: Fractions: multiplying improper fractions and mixed numbers

Syllabus outcomes: Fractions 2.3, Fractions 2.4, Fractions 2.5. Notes on the tag: multiplying a proper fraction by an improper fraction, two improper fractions, a mixed number by a whole number
Puzzle type: Lettered choices (fraction products). How the pupil checks: Word from the letters of the right choices (6 letters).
Answers: 1) 2 1/12; 2) 14 1/4; 3) 9 1/6; 4) 1 2/7; 5) 6 2/5; 6) 1 13/15
Check result: Word: CREAMY
Teacher note: Six products: two improper fractions times each other (two rows), a proper fraction times an improper fraction, an improper fraction times a whole number and a mixed number times a whole number (two rows). The wrong choices are the usual slips (adding the denominators, adding the numerators, not multiplying the fraction part of a mixed number, or answers one whole or one half away). Each choice has a letter and the right choices spell the word. Every other way of choosing was tested against a very large word list. A box left blank gives no letter and the check cannot see it.

S1-16 (P5-16) Remainder Stories 2 stars

Topic: Fractions: a fraction of the remainder

Syllabus outcomes: Fractions 2.2, Fractions 2.3 (problem solving). Notes on the tag: multiplying a fraction by a whole number: a fraction of a quantity, then a fraction of what is left
Puzzle type: Code-breaker (two-step fraction stories). How the pupil checks: Code word (6 letters).
Answers: 1) 4; 2) 9; 3) 18; 4) 19; 5) 8; 6) 3
Check result: Code word: FAMILY
Teacher note: Six two-step stories. The pupil takes a fraction of the starting amount, then a fraction of what is left, and answers one of three questions: what is left, what was taken the second time, or what was taken in all. The decoy letters stand for the usual slips: taking the second fraction of the whole instead of the rest, or answering a different one of the three questions. Every step divides exactly. A box left blank gives no letter and the check cannot see it.

S1-17 (P5-17) Point Shift 1 star

Topic: Decimals: multiplying and dividing by 10, 100 and 1000

Syllabus outcomes: Decimals 1.1. Notes on the tag: multiplying and dividing decimals (up to 3 decimal places) by 10, 100 and 1000
Puzzle type: Code-breaker (decimal point shifts). How the pupil checks: Code word (6 letters).
Answers: 1) 0.37; 2) 5510; 3) 56; 4) 0.9; 5) 0.06; 6) 0.688
Check result: Code word: POLICE
Teacher note: Six calculations: a whole number or a decimal of up to three places multiplied or divided by 10, 100 or 1000. No answer has four or more decimal places. The decoy letters stand for the usual slips: the decimal point moved one place too many or too few, or moved the wrong way. A box left blank gives no letter and the check cannot see it.

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S1-18 (P5-18) True or False 2 stars

Topic: Decimals: multiplying and dividing by 10, 100, 1000 and their multiples

Syllabus outcomes: Decimals 1.1. Notes on the tag: multiplying and dividing decimals by 10, 100, 1000 and their multiples
Puzzle type: True or false table. How the pupil checks: Word from the letters of the right choices (6 letters).
Answers: 1) F (5.9 × 600 = 35 400); 2) T (45 ÷ 10 = 4.5); 3) T (84.9 ÷ 10 = 8.49); 4) T (0.77 × 2000 = 1540); 5) F (885 ÷ 10 = 8.85); 6) T (9.1 × 600 = 5460)
Check result: Word: FIGURE
Teacher note: Six statements with decimals multiplied or divided by 10, 100, 1000 or a multiple of 10, 100 or 1000. Each row has a T choice and an F choice with a letter. The false statements are the usual slips (the decimal point one place out, or the wrong number of zeros). The letters of the right choices spell the word. Every other way of choosing was tested against a very large word list. A box left blank gives no letter and the check cannot see it.

S1-19 (P5-19) Measures in Decimals 1 star

Topic: Decimals: converting measures to a larger or smaller unit

Syllabus outcomes: Decimals 1.2. Notes on the tag: converting a measure to a larger or smaller unit in decimal form (km and m, m and cm, kg and g, ℓ and mℓ)
Puzzle type: Lettered choices (units). How the pupil checks: Word from the letters of the right choices (6 letters).
Answers: 1) 7.944 ℓ; 2) 3.26 m; 3) 8.615 kg; 4) 70 m; 5) 86 g; 6) 11 mℓ
Check result: Word: HUNGRY
Teacher note: Six conversions in km and m, m and cm, kg and g, and ℓ and mℓ: three from two units to a decimal in the larger unit and three from a decimal to the smaller unit. The wrong choices are the usual slips (the point one or two places out, the digits of the smaller unit put in the wrong places). Each choice has a letter and the right choices spell the word. Every other way of choosing was tested against a very large word list. A box left blank gives no letter and the check cannot see it.

S1-20 (P5-20) Unit Switch 2 stars

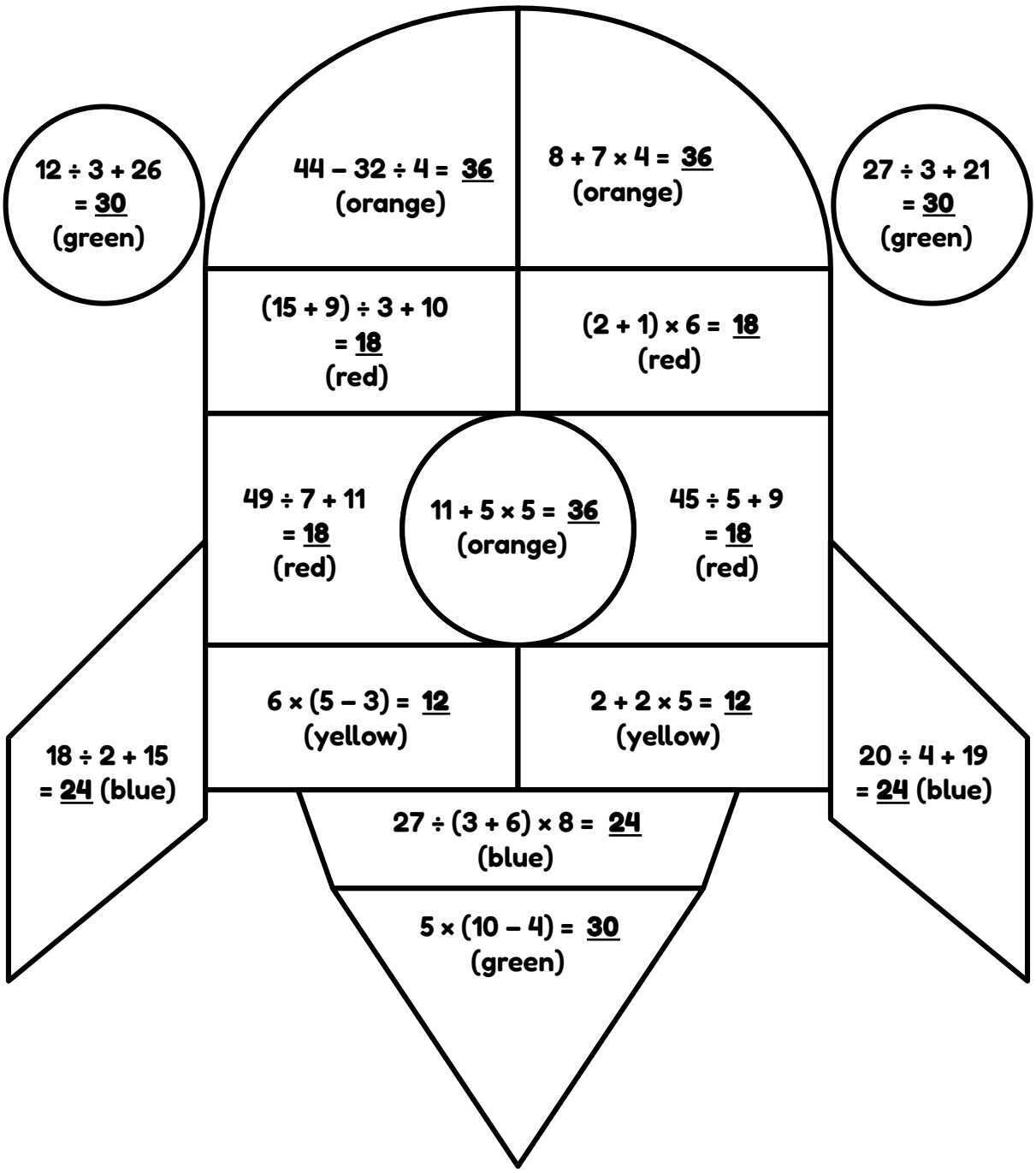
Topic: Decimals: converting measures and solving problems

Syllabus outcomes: Decimals 1.2 (problem solving). Notes on the tag: converting a measure to a larger or smaller unit in decimal form, then solving a short problem
Puzzle type: Code-breaker (measures in stories). How the pupil checks: Code word (6 letters).
Answers: 1) 2750 mℓ; 2) 333 cm; 3) 4980 m; 4) 3206 g; 5) 2034 mℓ; 6) 30 cm
Check result: Code word: FRUITS
Teacher note: Six short stories with decimal measures (ℓ, m, km, kg): the pupil changes one measure to the smaller unit and then multiplies, adds, subtracts or divides. The decoy letters stand for the usual slips: the unit changed by the wrong power of ten. All answers are whole numbers of the unit shown. A box left blank gives no letter and the check cannot see it.

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Answers: S1-05 (P5-05) Colour the Rocket

Every part shows the question, the answer underlined and the colour in brackets. Pupils colour their own copy. Colour key: 12 = yellow, 18 = red, 24 = blue, 30 = green, 36 = orange.



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